

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC Engineering Services

Subject: Electrical Engineering

Questions: 141

Source: Official source PDF extraction

AI-generated PYQs: 0

1. [UNKNOWN] unknown

1. What are the values of k for which the system of equations $(3k-8)x + 3y + 3z = 0$, $3x + (3k-8)y + 3z = 0$, $3x + 3y + (3k-8)z = 0$ has a non-trivial solution? $k = -2, -11, -10, -3$ (d) $-3, -11, -10, -2$
The solution of the differential equation $x \frac{d^2y}{dx^2} + y \log x$ is $y = (c_1 + c_2x) \log x + 2 \log x + 3y = (c_1 + c_2x^2) \log x + \log x + 2y = (c_1 - c_2 \log x)x + \log x + 2$

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2. [UNKNOWN] unknown

1. $1, xyz, 0, x, y, 1 \pm 1 + 1, xy, 0, x, y, z, z, -1, (\cos a + i \sin a)^2, 4, 4, -1, (\cos LC, 1 - i \sin Lt)^2, 2, 2, 2, (C)^2, (d)^2, -1, \cos, -it + i \sin, -2, 6, 6$

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3. [UNKNOWN] unknown

1. $j, n, 1, -j, n, 1, -j, n, 1, (e) = e + -e$ and no $--2, 2, 6, (d) = -1, e, i, n, 0, n + -1, e, n, n, 0, n$ and $Q_0 = I, t, 2, 2, 4$

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4. [UNKNOWN] unknown

2. If $2+1, 3, -1+3i, A = -5, i, 4, -2i$ then AA^* will be (where, A is the conjugate transpose of A) Unitary matrix Orthogonal matrix Hermitian matrix Skew Hermitian matrix

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5. [UNKNOWN] unknown

3. If $y = -3x^2 + 3x - 10$, the value of A^3y will be (where, A is forward differences operator) (a) 10 (b) 11 (c) 12 (d) 13
 $16, 2, -a, 3, 7, 1, -a, 2, 3$

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6. [UNKNOWN] unknown

5. The area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is (a) $-2a^2, 3$ (b) $1a^2, 3$

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7. [UNKNOWN] unknown

6. The volume of the solid surrounded by the surface $(x)^{2/3} + (y)^{2/3} + (z)^{2/3} = 1$ (a) (13) (c) $4\pi abc$ $35abc$ $35abc$ $35itabc$ 35
VBYS-ETE (2 - A) ===== PAGE 3 =====

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8. [UNKNOWN] unknown

7. The solution of the partial differential equation $x^2 \frac{\partial^2 f}{\partial x^2} - a \frac{\partial f}{\partial x} + y^2 \frac{\partial^2 f}{\partial y^2} - b \frac{\partial f}{\partial y} = f$ is $f(x, y) = \frac{1}{x^2} \log \frac{y}{x} + \frac{1}{y^2} \log \frac{x}{y}$

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9. [UNKNOWN] unknown

9. If n is a positive integer then, $(1 + R)^n = 1 + nR + \frac{n(n-1)}{2}R^2 + \dots + R^n$

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10. [UNKNOWN] unknown

10. The nature of singularity of function $f(z) = \frac{1}{z} \cos z$ at $z = 0$ is $\cos z = \sin z$ 4 Removable singularity Isolated singularity

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11. [UNKNOWN] unknown

10) = $3 \cos 100 \text{ nt}$. The minimum sampling rate F_s required to avoid aliasing will be 100 Hz 200 Hz 300 Hz 400 Hz

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12. [UNKNOWN] unknown

11. If X is a discrete random variable that follows Binomial distribution, then which one of the following response relations is correct? $P(r + 1) - n - rP(r)$ $r + 1 P(r + 1) = P(r)$ $P(r + 1) = n r P(r)$ $r + 1 q P(r + 1) = n - rP(r)$ $r + 1 q$

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13. [UNKNOWN] unknown

12. If the probability that an individual suffers a bad reaction from a certain infection is 0.001, what is the probability that out of 2000 individuals, more than 2 individuals will suffer a bad reaction? (a) 1.5×10^{-5} (b) 1.2×10^{-5} (c) 5×10^{-5} (d) 5×10^{-6}

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14. [UNKNOWN] unknown

13. Materials in which the atomic order extends uninterrupted over the entirety of the specimen; under some may have flat faces shapes, are called Anisotropy Crystallography Single crystals Crystal system circumstances, they regular geometric and

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15. [UNKNOWN] unknown

14. Which material possesses the following properties? Shining white colour with lustre Soft, malleable and can be drawn into wires Poor in conductivity and tensile strength Used in making alloys with lead and copper Used for fuses and cable sheathing Silver Tin Nickel Aluminium

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16. [UNKNOWN] unknown

15. The saturation magnetization for Fe_3O_4 , given that each cubic unit cell contains 8 Fe^{2+} and 16 Fe^{3+} ions, where Bohr magneton is $9.274 \times 10^{-24} \text{ A.m}^2$ and that the unit cell edge length is 0.839 nm, will be $1.25 \times 10^5 \text{ Aim}$ $5 \times 10^5 \text{ Aim}$ $10 \times 10^5 \text{ Aim}$ $20 \times 10^5 \text{ Aim}$

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17. [UNKNOWN] unknown

16. Consider the following applications of the materials: Bismuth strontium calcium copper oxide used as a high temperature superconductor Boron carbide used in helicopter and tank armour Uranium oxide used as fuel in nuclear reactors Bricks used for construction The materials used in these applications can be classified as Ceramic Constantan Manganin Tantalum

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18. [UNKNOWN] unknown

17. The saturation flux density for Nickel having density of 8.90 g/cm^3 , atomic number 58-71 and net magnetic moment per atom of 0.6 Bohr magnetons is nearly

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19. [UNKNOWN] unknown

18. The temperature at which iron ceases to be ferromagnetic and becomes paramagnetic is Curie-Weiss point Thermo-magnetic point Ferro-paramagnetic point Curie point

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20. [UNKNOWN] unknown

19. Fick's laws refer to Finding whether a semiconductor is n or p type Diffusion Crystal imperfections Electric breakdown

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21. [UNKNOWN] unknown

20. A magnetic field applied perpendicular to the direction of motion of a charged particle exerts a force on the particle perpendicular to both the magnetic field and the direction of motion of the particle. This phenomenon results in Flux effect Hall Effect Magnetic field effect Field effect

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22. [UNKNOWN] unknown

21. An electric kettle is marked 500 W, 230 V and is found to take 15 minutes to bring 1 kg of water at 15°C to 100°C. If the specific heat of water is 4200 J/kg/°C, the heat efficiency of the kettle will be 87.3% 83.6% 79.3% 75.6% 0.82 tesla 0.76 tesla 0.64 tesla 0.52 tesla VBY-S-ETE (4-A) ===== PAGE 5 =====

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23. [UNKNOWN] unknown

22. With reference to nano materials, the prefix nano stands for Nano centimetre Nanometre Nano micrometre Nano millimetre

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24. [UNKNOWN] unknown

23. Consider the following applications: High temperature heat engines Nuclear fusion reactors Chemical processing industry Aeronautical and space industry Which one of the following materials will be used for these applications? Zirconia Alumina Ceramic Silicon carbide

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25. [UNKNOWN] unknown

24. The machine used for the preparation of nano particles of alumina is Attrition mill Grinding machine Vending machine Welding machine

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26. [UNKNOWN] unknown

25. If the voltage across an element in a circuit is linearly proportional to the current through it, then it is a Capacitor Transformer Resistor Inductor

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27. [UNKNOWN] unknown

26. Thevenin's equivalent circuit consists of Current source and series impedance Voltage source and series impedance Voltage source and shunt impedance Current source and shunt impedance

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28. [UNKNOWN] unknown

27. When the voltage sources are replaced with short circuits and current sources are replaced with open circuits, leaving dependent sources in the circuit, the theorem applied is Superposition Thevenin Norton Millman

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29. [UNKNOWN] unknown

28. The maximum power is delivered from a source to a load when the source resistance is Greater than the load resistance Equal to zero Less than the load resistance Equal to the load resistance

30. [UNKNOWN] unknown

29. A network delivers maximum power to the load resistance when it is Greater than Norton's equivalent resistance of the network Equal to Thevenin's equivalent resistance of the network Less than source resistance Less than Norton's equivalent resistance of the network

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31. [UNKNOWN] unknown

30. The impedance of a parallel circuit is $(10 - j 30) \Omega$ at 1 MHz. The values of circuit elements will be 10Ω and 6.4 mH 100Ω and 4.7 nF 10Ω and 4.7 mH 100Ω and 6.4 nF VBY-S-ETE (5—A) ===== PAGE 6 =====

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32. [UNKNOWN] unknown

31. The defining equations for V_1 and V_2 analyzing a two-port network in terms of its impedance parameters are respectively $Z_{11}I_1 + Z_{12}I_2$ and $Z_{21}I_1 + Z_{22}I_2$ $Z_{11}I_1 + Z_{12}I_2$ and $Z_{21}I_1 + Z_{22}I_2$ (c). $Z_{11}I_1 + Z_{21}I_2$ and $Z_{12}I_1 + Z_{22}I_2$ (d) $Z_{11}I_1 + Z_{12}I_2$ and $Z_{21}I_1 + Z_{22}I_2$

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33. [UNKNOWN] unknown

32. A filter that allows high and low frequencies to pass but attenuates any signal with a frequency between two corner frequencies is a Notch filter Band pass filter Band stop filter Multiband filter

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34. [UNKNOWN] unknown

33. When a number of two-port networks are cascaded then z-parameters are added up y-parameters are added up h-parameters are multiplied ABCD-parameters are multiplied

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35. [UNKNOWN] unknown

34. A 3-phase star-connected 1000 volt alternator supplies power to a 500 kW delta-connected induction motor. If the motor power factor. is 0.8 lagging and its efficiency 0.9, then the current in each alternator and motor phase respectively are nearly 321 A and 231.5 A 401 A and 231.5 A 321 A and 1854A 401 A and 185.4 A

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36. [UNKNOWN] unknown

35. Consider the following statements • Mutual inductance describes the voltage induced at the ends of a coil due to the magnetic field generated by a second coil. The dot convention allows a sign to be assigned to the voltage induced due to mutual inductance term. The coupling coefficient is given by $k = \frac{M}{\sqrt{L_1 L_2}}$ Which of the above statements are correct? 1, 2 and 3 1 and 3 only 1 and 2 only 2 and 3 only

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37. [UNKNOWN] unknown

36. Consider the following statements: The rules for series and parallel combinations of capacitors are opposite to those for resistors. The rules for series and parallel combinations of inductors are same as those for resistors. An inductor is a short circuit to dc currents Which of the above statements are correct? 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3 The standard resistor is a coil of wire of some alloy having the properties of Low electrical resistivity and high temperature coefficient of resistance High electrical resistivity and high temperature coefficient of resistance Low electrical resistivity and low temperature coefficient of resistance High electrical resistivity and low temperature coefficient of resistance

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38. [UNKNOWN] unknown

37. VBY-S-ETE (6—A) ===== PAGE 7 =====

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39. [UNKNOWN] unknown

38. Which one of the following materials is used for the swamping resistance of moving coil instruments? Carbon
Manganin Silver Brass

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40. [UNKNOWN] unknown

39. In a PMMC instrument, the swamping resistor is used to Increase the damping of the instrument Reduce the current
within safe limits Compensate for temperature variations Increase the full-scale sensitivity

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41. [UNKNOWN] unknown

40. A moving coil ammeter has a fixed shunt of 0.02 a With a coil resistance of $R = 1000$ and a potential difference of
500 mV across it, full scale deflection is obtained The current through the moving coil to give full scale deflection will be
25A 0.5×10^{-2} A 0.25×10^{-3} A 0.5×10^{-3} A

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42. [UNKNOWN] unknown

41. A moving iron instrument has full scale current of 100 mA. It is converted into a 250 V voltmeter by using a series
resistance made of a material having negligible resistance temperature coefficient. The meter has a resistance of 320 Ω
at 20°C. After carrying a steady current of 100 mA for a long time, the resistance of the coil increases to 369 Ω due to
self heating. When a voltage of 250 V is applied continuously, the error due to self-heating will be nearly -11% - 1.9%
-25% - 3.3%

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43. [UNKNOWN] unknown

42. There will be serious errors if power factor of non-sinusoidal waveform is measured by electro-dynamometer power
factor meter. This is true for Single-phase meters alone 3-phase meters only Both Single-phase meters and 3-phase
meters 3-phase meters with balanced loads

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44. [UNKNOWN] unknown

43. The ramp type digital voltmeter can measure accurately with A positive going ramp voltage only A negative or
positive going linear ramp voltage A negative going ramp voltage only An asymptotic ramp voltage only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

45. [UNKNOWN] unknown

44. The self-capacitance of a coil is measured by the resonating capacitor. The measurement gives the value of tuning
capacitor as $C_1 = 460$ pF at a frequency, $f_1 = 2$ MHz. The second measurement at $f_2 = 4$ MHz yields a new value of
tuning capacitor, $C_2 = 100$ pF. The distributed capacitance C_d will be 10 pF 20 pF 30 pF 40 pF

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46. [UNKNOWN] unknown

45. Vertical delay line in CRO Gives proper time for thermionic emission of electrons Delays the signal voltage by 200 ns
Allows the horizontal sweep to start prior to vertical deflection Delays the generation of sweep voltage VBY-S-ETE (7-A)
===== PAGE 8 =====

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47. [UNKNOWN] unknown

46. A 0 — 150 V voltmeter has a guaranteed 51. accuracy of 1% full scale reading. The voltage measured by this instrument is 83 V. The limiting error will be nearly 1.2% 1.8% 2-4% 3.2% Convert decimal 41.6875 into equivalent binary: 100101.1011 100101.1101 1010014011 101001.1101

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48. [UNKNOWN] unknown

47. The variations in the measured quantity due to sensitivity of transducer to any plane other than the required plane is Cross sensitivity Sensitivity Interference Distributed sensitivity

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49. [UNKNOWN] unknown

48. A resistance strain gauge with a gauge factor of 2 is fastened to a steel member subjected to a stress of 1050 kg/cm². The modulus of elasticity of steel is 2.1×10^6 kg/cm². The change in resistance AR of the strain gauge element due to the applied stress will be 04% 0.2% 0-3% 0.4%

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50. [UNKNOWN] unknown

49. In which one of the following classes of computers, is the relationship between architecture and organization very close? Microcomputers Mini computers Mainframe computers Super computers

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51. [UNKNOWN] unknown

50. The decimal equivalent of binary number 1001401 is 9-750 W625 10750 10625

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52. [UNKNOWN] unknown

52. The Central Processing Unit (CPU) consists of ALU and Control unit only ALU, Control unit and Registers only ALU, Control unit and System bus only ALU, Control unit, Registers and Internal bus

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53. [UNKNOWN] unknown

53. When enough total memory space exists to satisfy a request, but it is not contiguous, then this problem is known as Internal Fragmentation External Fragmentation Overlays Partitioning

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54. [UNKNOWN] unknown

54. The total average read or write time T_{tstai} is $T_s + \frac{1}{2} + \frac{b}{2rN} T_s + \frac{1}{2} + \frac{b}{2rN} T_s$ $b \frac{1}{2} + \frac{b}{2rN} T_s$ $b \frac{1}{2} + \frac{b}{2rN} T_s + 2r + \frac{b}{2rN}$ rN where, T_s = average seek time b = number of bytes to be transferred N = number of bytes on a track r = rotation speed, in revolutions per second VBY-S-ETE (8—A) ===== PAGE 9 =====

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55. [UNKNOWN] unknown

55. If a cache has 64-byte cache lines, how long does it take to fetch a cache line if the main memory takes 20 cycles to respond to each memory request and returns 2 bytes of data in response to each request? 980 cycles 640 cycles 320 cycles 160 cycles

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56. [UNKNOWN] unknown

56. Which of the following statements are correct about SRAM? It provides faster access as compared to DRAM. It is cheaper than DRAM. It is more expensive than DRAM. It has higher bit density than DRAM. 1 and 4 only 1 and 3 only 1, 3 and 4 only 2 and 4 only

57. [UNKNOWN] unknown

57. Features of solid state drives (SSDs) are High-performance in input/output operations per second More power consumption than comparable size HDDs Lower access times and latency rates More susceptible to physical shock and vibration 2 and 3 only 2 and 4 only 1 and 3 only 1 and 4 only

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58. [UNKNOWN] unknown

58. The decimal value of signed binary number 11101000 expressed in I's complement is — 223 — 184 — 104 — 23

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59. [UNKNOWN] unknown

59. The memory management function of virtual memory includes Space allocation Program relocation Program execution Code sharing 1, 2 and 3 only 1, 2 and 4 only 1, 3 and 4 only 2, 3 and 4 only

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60. [UNKNOWN] unknown

60. Which of the following instructions of 8085 are the examples of implied addressing? CMA IN byte RET 1, 2 and 3 1 and 2 only 2 and 3 only 1 and 3 only VBY-S-ETE (9—A) ===== PAGE 10 =====

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61. [UNKNOWN] unknown

61. The important fact about the collector current is (a) It is greater than emitter current It equals the base current divided by the current gain It is small It approximately equals• the emitter current

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62. [UNKNOWN] unknown

62. What is Shockley's equation of a semiconductor diode in the forward bias regions? (a) $I_D = I_s (e^{V_D / nV_T} - 1)$ $I_D = I_s (e^{-1})$ $I_D = I_s (e^{-1})$ $I_D = I_s (e^{-1})$ 17T1 nVD ID = Is (e —1) where Is is reverse saturation current VD is applied forward-bias voltage across the diode VT is thermal voltage n is an ideality factor

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63. [UNKNOWN] unknown

63. The thermal voltage VT of a semiconductor diode at 27°C temperature is nearly 17 mV 20 mV 23 mV 26 mV

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64. [UNKNOWN] unknown

64. The disadvantage of a typical MOSFET as compared to BJT is Increased power-handling levels Reduced power-handling levels Increased voltage-handling levels Reduced voltage-handling levels

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65. [UNKNOWN] unknown

65. Which one of the following conditions will be satisfied for an impedance matched system? The decibel power gain is equal to twice the decibel voltage gain The decibel power gain is equal to the decibel voltage gain The decibel power gain is half the decibel voltage gain The decibel power gain is equal to thrice the decibel voltage gain

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66. [UNKNOWN] unknown

66. For most FET configurations and for common-gate configurations, the input impedances are respectively High and high High and low Low and low Low and high

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67. [UNKNOWN] unknown

67. The dB gain of cascaded systems is simply The square of the dB gains of each stage The sum of the dB gains of each stage The multiplication of the (113 gains of each stage The division of the dB gains of each stage

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68. [UNKNOWN] unknown

68. The Miller effect input capacitance C_{mi} is $(1 - AV_2) C_f$ $(1 - A_v) C_f$ $(1 - C_f) A_v$ $(1 - C_f^2) A_v$ where C_f = feedback capacitance $V A_v$

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69. [UNKNOWN] unknown

69. For an op-amp having a slew rate of 2 V/ps, if the input signal varies by 0-5 V in 10 ms, the maximum closed-loop voltage gain will be 50 40 22 20 V/V-S/ET (10 — A) ===== PAGE 11 =====

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70. [UNKNOWN] unknown

70. A negative feedback amplifier where an input current controls an output voltage is called Current amplifier Transconductance amplifier Transresistance amplifier Voltage amplifier

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71. [UNKNOWN] unknown

71. In emergency lighting system, the component used for maintaining the charge on the battery is LED Shockley diode Thermistor SCR

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

72. [UNKNOWN] unknown

72. For RC phase shift oscillator using FET, the gain of the amplifier stage must be practically somewhat greater than 27 28 29 30

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

73. [UNKNOWN] unknown

73. The time delay in a look-ahead carry adder is independent of Number of operands only Propagation delay only Number of bits in the operand only Bits in the operand, number of operands and propagation delay

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

74. [UNKNOWN] unknown

74. The time delay A_t introduced by a SISO shift register in digital signals is given by • (a)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

75. [UNKNOWN] unknown

75. An analog output voltage for the input 1001 to a 4 bit D/A converter for all possible inputs assuming the proportionality factor $K = 1$ will be 9 6 3 1

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

76. [UNKNOWN] unknown

76. In microprocessor interface, the concept of detecting some error condition as 'no match found' is called Syntax error Semantic error Logical error Error trapping

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

77. [UNKNOWN] unknown

77. The maximum number of input or output devices that can be connected to 8085 microprocessor are 8 16 40 256

78. [UNKNOWN] unknown

78. The contents of the accumulator and register C are 2EH and 6CH respectively. The instruction ADD C is used. The values of AC and P flags are 0 and 0 1 and 1 0 and 1 1 and 0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

79. [UNKNOWN] unknown

79. When an information signal is multiplied by an auxiliary sinusoidal signal to translate its frequency, the modulation is called Phase modulation Frequency modulation Amplitude modulation Quadrature amplitude modulation $N^2 \times f_c$ (d) $N \times f$, where N is the number of stages f, is the clock frequency such VBY-S-ETE (11 — A) ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

80. [UNKNOWN] unknown

80. The transmission power efficiency for a tone modulated signal with modulated index of 0.5 will be nearly (a).. (b) (b) 11.1% (c) 16.7% (d) 21.1%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

81. [UNKNOWN] unknown

81. For practical purposes, the signal to noise ratio for acceptable quality transmission of analog signals and digital signals respectively are (a) 10 — 30 dB and 05 — 08 dB (b) 40 — 60 dB and 10 — 12 dB (c) 60 — 80 dB and 20 — 24 dB (d) 70 — 90 dB and 30 — 36 dB

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

82. [UNKNOWN] unknown

82. The discrete samples of an analog signal are to be uniformly quantized for PCM system. If the maximum value of the analog sample is to be represented within 04% accuracy, then minimum number of binary digits required will be nearly 7 9 11 13

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

83. [UNKNOWN] unknown

83. A signal : $m(t) = 2 \cos 6000 \pi t + 4 \cos 8000 \pi t + 6 \cos 10000 \pi t$ is to be truthfully represented by its samples. The minimum sampling rate using band pass consideration will be 5,000 Hz 10,000 Hz 15,000 Hz 20,000 Hz

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

84. [UNKNOWN] unknown

84. If 'IT signals are multiplexed using PAM band limited to f_k , the channel bandwidth need not be larger than $f_{,,}$, $N \cdot a$ 2 $N \cdot f_m$ 2N $\cdot f$ (d) $N^2 \cdot f_m$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

85. [UNKNOWN] unknown

85. A linear discrete-time system is characterized by its response $h_k(n) = (n - k)u(n - k)$ to a delayed unit sample $\delta(n - k)$. The system will be Shift invariant Shift variant Scale invariant Scale variant

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

86. [UNKNOWN] unknown

87. The response of the system $y(n) = x(n)$ to the following input signal $x(n) = 1$ for $n \geq 0$, otherwise 0, is delayed from input 1s exactly same as the input Leads the input Varies with signal VI3Y-S-ETE (12 — A) ===== PAGE 13 ===== The complex exponential Fourier representation for the signal $x(t) = \cos \omega t$ is ,k

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

87. [UNKNOWN] unknown

89. The continuous LTI system is described by $dy(t) + 2y(t) = x(t)$. Using the Fourier transform, for $x(t) = e^{-t} u(t)$, the output $y(t)$ will be $(e^{-t} - e^{-2t}) u(t)$. The discrete Fourier series representation for the following sequence $x(n) = \cos \left(\frac{\pi}{4} n \right)$ is $\frac{1}{2} e^{j \frac{\pi}{4} n} + \frac{1}{2} e^{-j \frac{\pi}{4} n}$. (a) $\frac{1}{2} e^{j \frac{\pi}{4} n} + \frac{1}{2} e^{-j \frac{\pi}{4} n}$ and $\frac{1}{2} e^{j \frac{\pi}{4} n} - \frac{1}{2} e^{-j \frac{\pi}{4} n}$ (b) $\frac{1}{2} e^{j \frac{\pi}{4} n} - \frac{1}{2} e^{-j \frac{\pi}{4} n}$ and $\frac{1}{2} e^{j \frac{\pi}{4} n} + \frac{1}{2} e^{-j \frac{\pi}{4} n}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

88. [UNKNOWN] unknown

91. Consider the discrete-time sequence $x(n) = \cos \left(\frac{\pi}{8} n \right)$. When sampled at frequency $f_s = 10$ kHz, then f_0 , the frequency of the sampled continuous time signal which produced this sequence will at least be 625 Hz, 575 Hz, 525 Hz, 475 Hz.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

89. [UNKNOWN] unknown

92. How many bits are required in an A/D converter with a $B+1$ quantizer to get a signal-to-quantization noise ratio of at least 90 dB for a Gaussian signal with range of $\pm 3\sigma$? $B+1 = 12$ bits, $B+1 = 14$ bits, $B+1 = 15$ bits, $B+1 = 16$ bits.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

90. [UNKNOWN] unknown

93. Let $x(n)$ be a left-sided sequence that is equal to $3z^{-1} + 2z^{-2}$. Then $x(0)$ will be 0, 2, 3, 4.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

91. [UNKNOWN] unknown

94. The noise variance σ^2 at the output of $0.5z^{-1} H(z)$ with respect to input will be $\sigma^2 = 0.6$, nearly 40%, 50%, 60%, 70%. (a) to zero for $n > 0$. If $X(z) = \frac{1}{1 - 0.5z^{-1}}$, then $x(n)$ is $(0.5)^n u(n)$. (d) $K v$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

92. [UNKNOWN] unknown

95. If the complex multiply operation takes 1 μ s, the time taken to compute 1024-point DFT directly will be nearly 3.45 s, 2.30 s, 1.05 s, 0.60 s.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

93. [UNKNOWN] unknown

96. Consider the following data to design a low-pass filter. Cut-off frequency $\omega_c = \frac{\pi}{2}$. Stop band ripple $\delta_s = 0.002$, Transition bandwidth no larger than 0.1 rad. Kaiser window parameters β and N respectively are 2.99 and 45, 4.99 and 45, 2.99 and 65, 4.99 and 65.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

94. [UNKNOWN] unknown

97. A transfer function $G(s) = \frac{1}{1 + sT}$ has a phase $\angle G(j\omega)$ which varies from 0° to -180° as ω is increased from 0 to ∞ . It is the transfer function for All pass system, Low pass system, High pass system, Band pass system.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

95. [UNKNOWN] unknown

98. The open-loop and closed-loop transfer functions of a system are respectively given by

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

96. [UNKNOWN] unknown

99. The system sensitivity of open loop and closed loop system are respectively $\frac{1}{1 + GH}$ and $\frac{1}{1 + GH}$, $\frac{1}{1 + GH}$ and $\frac{1}{1 + GH}$, $\frac{1}{1 + GH}$ and $\frac{1}{1 + GH}$, $\frac{1}{1 + GH}$ and $\frac{1}{1 + GH}$.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

97. [UNKNOWN] unknown

100. The steady state error of a type-1 system to a unit step input is

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

98. [UNKNOWN] unknown

101. The direction of the net encirclements of the origin of Real-Imaginary plane in a Nyquist plot for the system to be stable is Clockwise of the origin Counter-Clockwise of the origin Left hand side s-plane Right hand side s-plane $G(s) = \frac{1}{s^2 + 1}$; (open loop), $(1 + K) G(s) = \frac{1}{s^2 + 1}$; (closed loop). j CO Te + 1 The ratio of the bandwidth of closed loop to open loop system is $K(1 + K)$ (c) $(1 + \frac{1}{K})$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

99. [UNKNOWN] unknown

102. A unity negative feedback control system has an open-loop transfer function as $K(s + 1)(s + 2)(s + 0.1)(s - 1)$ The range of values of K for which the closed loop system is stable will be $0 < K < 0.3$ $K > 0.3$ $K > 3$ $K < 0.3(1 + K)$
VBYS-ETE (14 - A) ===== PAGE 15 ===== $100(s + 60)(s + 10)(s + 60)(s + 10)600(s + 60)(s + 10)60(s + 60)(s + 10)10$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

100. [UNKNOWN] unknown

103. The lag system of a 'lag-lead compensator' has one pole and one zero. Then pole and zero are Real and pole is to the left of zero Real and pole is to the right of zero Imaginary and pole is above zero Imaginary and pole is below zero

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

101. [UNKNOWN] unknown

104. A system with characteristic equation $F(s) = s^4 + 6s^3 + 23s^2 + 40s + 50$ will have closed loop poles such that All poles lie in the left half of the s-plane and no pole lies on imaginary axis Two poles lie symmetrically on the imaginary axis of the s-plane All four poles lie on the imaginary axis of the s-plane All four poles lie in the right half of the s-plane

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

102. [UNKNOWN] unknown

105. A unity feedback (negative) system has open loop transfer function $s(s + 2)$ The closed loop system has a steady-state unit ramp error of 0.1. The value of gain K should be 20 30 40 50

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

103. [UNKNOWN] unknown

106. Transfer function of discrete time system derived from state model is given by $C(zI - A)^{-1}B + D$ $C(zI - A)^{-1}D + B$ $B(zI - A)^{-1}D + C$ $D(zI - A)^{-1}B + C$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

104. [UNKNOWN] unknown

107. The closed-loop response of a system subjected to a unit step input is $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$. The expression for the closed loop transfer function is

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

105. [UNKNOWN] unknown

108. If it is possible to transfer the system state $x(t_0)$ to any desired state $x(t)$ in specified finite time by a control vector $u(t)$, then the system is said to be Completely observable Completely state controllable Random state system Steady state controlled system

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

106. [UNKNOWN] unknown

109. Consider the following statements regarding parallel connection of 3-phase transformers : The secondaries of all

transformers must have the same phase sequence. The phase displacement between primary and secondary line voltages must be the same for all transformers which are to be operated in parallel. The primaries of all transformers must have the same magnitude of line voltage. Which of the above statements are correct? 1, 2 and 3 1 and 3 only 1 and 2 only 2 and 3 only VBY-S-ETE

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

107. [UNKNOWN] unknown

110. A 500 kVA transformer has an efficiency of 95% at load and also at 60% of full load, both at upf. The efficiency of the transformer at $\frac{1}{3}$ th full load will be nearly 4 98% 95% 92% 87%

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

108. [UNKNOWN] unknown

111. What is the condition of retrogressive winding in dc machines?

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

109. [UNKNOWN] unknown

112. What is the useful flux per pole on no load of a 250 V, 6-pole shunt motor having a wave connected armature winding with 110 turns, armature resistance of 0.2 Ω and armature current 13.3 A at no load speed of 908 rpm? 12.4 mWb 22.6 mWb 24.8 mWb 49.5 mWb

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

110. [UNKNOWN] unknown

113. The cross-magnetizing effect of the armature reaction can be reduced by Making pole shoes flat faced Making the main field ampere-turns larger compared to the armature-ampere turns Increasing the flux density under one half of the pole Keeping the direction of rotation of generator in the same direction as motor

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

111. [UNKNOWN] unknown

114. A 500 kW, 500 V, 10-pole, dc generator has a lap wound armature with 800 conductors. If the pole face covers 75% of pole pitch, the number of pole-face conductors in each pole of a compensating winding will be 12 10 8 6

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

112. [UNKNOWN] unknown

115. Cogging in an induction motor is caused If the number of stator slots are unequal to number of rotor slots If the number of stator slots are an integral multiple of rotor slots If the motor is running at fraction of its rated speed Due to 5th harmonic

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

113. [UNKNOWN] unknown

116. A 500 hp, 6-pole, 3-phase, 440 V, 50 Hz induction motor has a speed of 950 rpm on full-load. The full load slip and the number of cycles the rotor voltage makes per minute will be respectively 10% and 150 10% and 125 5% and 150 5% and 125

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

114. [UNKNOWN] unknown

117. Effective armature resistance $R_a(\text{eff})$ of a synchronous machine is Short circuit load loss (per phase) (Short circuit armature current)² Short circuit load loss (per phase) Short circuit load current Total short circuit load loss Short circuit armature current Total short circuit load loss Short circuit load current VBY-S-ETE (16—A) ===== PAGE 17 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

115. [UNKNOWN] unknown

118. A 3-phase synchronous motor has 12-poles and operates from 440 V, 50 Hz supply. If it takes a line current of

100 A at 0.8 power factor leading, its speed and torque are nearly 500 rpm and 1165 N-m 1000 rpm and 2330 N-m 500 rpm and 2330 N-m 1000 rpm and 1165 N-m

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

116. [UNKNOWN] unknown

119. Which of the following are the advantages of using a stepper motor? Compatibility with transformers and sensors needed for position sensing Compatibility with digital systems and sensors are not required for position and speed sensing Resonance effect often exhibited at low speeds and decreasing torque with increasing speed Easy to operate at high speeds and compatible with analog systems

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

117. [UNKNOWN] unknown

120. The disadvantage of hunting in synchronous machines is Fault occurs in the supply system Causes sudden change in inertia Causes large mechanical stresses and fatigue in the rotor shaft Causes harmonics

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

118. [UNKNOWN] unknown

121. Consider the following statements for a large national interconnected grid: Better load frequency control Same total installed capacity can meet lower demands Better hydro / thermal / nuclear / renewable co-ordination and energy conservation Which of the above statements are correct? 1 and 3 only 1 and 2 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

119. [UNKNOWN] unknown

122. A single-phase transformer is rated 110/440 V. 2.5 kVA. Leakage reactance measured from the low-tension side is 0.06 Q. The per unit leakage reactance will be 0.0062/unit 0.0124/unit 0.0496/unit 0.1983/unit

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

120. [UNKNOWN] unknown

123. A concentric cable has a conductor diameter of 1 cm and an insulation thickness of 1.5 cm. When the cable is subjected to a test pressure of 33 kV, the maximum field strength will be nearly 41,000 V 43,200 V 45,400 V 47,600 V

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

121. [UNKNOWN] unknown

124. Radio influence voltage (RIV) generated on a transmission line conductor surface is not affected by System voltage Corona discharges on the conductors Rain Nearby radio receivers VBY-S-ETE (17—A) ===== PAGE 18 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

122. [UNKNOWN] unknown

125. Consider the following properties regarding insulation for cables : A low specific resistance High temperature withstand High dielectric strength Which of the above properties of insulation are correct while using cables Z 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

123. [UNKNOWN] unknown

126. Which one of the following faults occurs more frequently in a power system? Grounded star-delta Double line faults LLG faults Single line-to-ground (LG) faults

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

124. [UNKNOWN] unknown

127. The maximum permissible time of de-energization of the faulty circuit is dependent on (a) Voltage of the system (h) The number of conductors involved Load carried by the faulty circuit Fault current and its duration

125. [UNKNOWN] unknown

128. Which one of the following is used for communication with the aim of achieving high figure of merit in HVDC circuit breakers? Oil interrupter Air interrupter Vacuum interrupter SF6 interrupter

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

126. [UNKNOWN] unknown

129. Which of the following buses are used to form bus admittance matrix for load flow analysis? Load bus Generator bus Slack bus 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

127. [UNKNOWN] unknown

130. In a 3-phase, 60 Hz, 500 MVA, 15 kV, 32-pole hydroelectric generating unit, the values of ω and ω_s will be nearly $\omega_s = 754 \text{ rad/s}$ and $\omega = 47.6 \text{ rad/s}$ $\omega_s = 377 \text{ rad/s}$ and $\omega = 46.7 \text{ rad/s}$ $\omega_s = 377 \text{ rad/s}$ and $\omega = 23.6 \text{ rad/s}$ $\omega_s = 754 \text{ rad/s}$ and $\omega = 23.6 \text{ rad/s}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

128. [UNKNOWN] unknown

131. The methods adopted for improving the steady state stability of power system are Quick response excitation system Higher excitation voltages Maximum power transfer by use of series capacitor or reactor 1 and 2 only 1 and 3 only 2 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

129. [UNKNOWN] unknown

132. The HVDC system uses Rectifier station at sending end and inverter station at receiving end Inverter station at sending as well as at the receiving end Rectifier station at sending end as well as at the receiving end Inverter station at sending end and rectifier station at receiving end

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

130. [UNKNOWN] unknown

133. Which one of the following is not required for Power diode? High speed operation Fast communication Small recovery time Low on-state voltage drop

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

131. [UNKNOWN] unknown

134. The reverse recovery time of a diode is $t_{rr} = 3 \mu\text{s}$ and the rate of fall of the diode current is $\frac{di}{dt} = 30 \text{ A/p.s.}$ The storage charge Q_{rr} and the peak inverse current I_{rr} will be respectively $135 \mu\text{C}$ and 90 A $270 \mu\text{C}$ and 90 A $270 \mu\text{C}$ and 60 A $135 \mu\text{C}$ and 60 A VBY-S-ETE (18 — A) ===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

132. [UNKNOWN] unknown

135. The $i_g - v_g$ characteristics of a thyristor is a straight line passing through origin with a gradient of 2.5×10^3 . If $P_g = 0.015 \text{ watt}$, the value of gate voltage will be nearly 5.0 V 6.1 V 75 V 8.5 V

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

133. [UNKNOWN] unknown

136. A single-phase 220 V, 1 kW heater is connected to half wave controlled rectifier and is fed from a 220 V, 50 Hz ac supply. When the firing angle $\alpha = 90^\circ$, the power absorbed by the heater will be nearly 1000 W 750 W 500 W 250 W

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

134. [UNKNOWN] unknown

137. When we compare the half bridge converter and full bridge converter The maximum collector current of a hill bridge

is only double that of the half bridge. Full bridge uses 4-power switches instead of 2, as in the double bridge. Output power of a full bridge is twice that of a half bridge with the same input voltage and current. Which of the above statements are correct? 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

135. [UNKNOWN] unknown

138. A single-phase hilly controlled bridge converter is connected with RLE load where $I_t = 5$ 1-2, $L = 4$ mH and $E = 50$ V. This converter circuit is supplied from 220 V, 50 Hz ac supply. When the firing angle is 60° , the average value of the load current will be nearly 12-2 A 9-8A 64A 4-2A

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

136. [UNKNOWN] unknown

139. Consider the following statements regarding ac drives: For the same kW rating, ac motors are 20% to 40% light weight as compared to de motors. The ac motors are more expensive as compared to same kW rating de motors. The ac motors have low maintenance as compared to dc motors. Which of the above statements are correct? 1 and 2 only 2 and 3 only 1 and 3 only 1, 2 and 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

137. [UNKNOWN] unknown

140. A 3-phase induction motor drives a blower where load torque is directly proportional to speed squared. If the motor operates at 1450 rpm, the maximum current in terms of rated current will be nearly 2-2 3.4 4.6 6.8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/ELECTRICAL-ENGINEERING-ESEP-19_0.pdf

138. [UNKNOWN] unknown

141. Consider the following statements: SMPS generates both the electromagnetic and radio frequency interference due to high switching frequency. SMPS has high ripple in output voltage and its regulation is poor. The output voltage of SMPS is less sensitive with respect to input voltage variation. Which of the above statements are correct? 1 and 3 only 2 and 3 only 1 and 2 only 1, 2 and 3

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139. [UNKNOWN] unknown

142. Consider the following features with respect to the flyback converters : It is used mostly in application below 100 W. It is widely used for high-output voltage. It has low cost and is simple. Which of the above statements are correct? 1, 2 and 3 1 and 2 only 1 and 3 only 2 and 3 only VBY-S-ETE (19 — A) ===== PAGE 20 =====

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140. [UNKNOWN] unknown

143. Consider the following statements regarding the function of dc-dc converter in a de motor: It acts as a regenerative brake. It controls the speed of motor. It controls the armature voltage of a dc motor. Which of the above statements are correct? 1 and 2 only 1 and 3 only 2 and 3 only 1,2 and 3

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141. [UNKNOWN] unknown

144. The power supplies which are used extensively in industrial applications are required to meet Isolation between the source and the load High conversion efficiency Low power density for reduction of size and weight Controlled direction of power flow Which of the above specifications are correct? 1, 2 and 3 only 1, 3 and 4 only 1, 2 and 4 only 2, 3 and 4 only Directions: Each of the next six (06) items consists of two statements, one labelled as 'Statement (I)' and the other as 'Statement (II)'. You are to examine these two statements carefully and select the answers to these items using the codes given below : Codes: Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I). Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I). Statement (I) is true, but Statement (II) is false. Statement (I) is false, but Statement (II) is true. Statement : Soft iron does not retain magnetism permanently. Statement (II) : Soft iron has no retentivity.

Statement (I) : Reaction turbines are generally used for sites with high head and low flow. Statement (II) : Kaplan and Francis turbines are reaction turbines. Statement (I) : One can formulate problems more efficiently in a high-level language and need not have a precise knowledge of the architecture of the computer. Statement (II) : High level languages permit programmers to describe tasks in a form which is problem oriented than computer oriented. Statement (I) : Sign magnitude representation is generally used in implementing the integer portion of the ALU. Statement (II) : In sign magnitude representation there are two representations of 0. Statement (I) : When a non-linear resistor, in series with a linear resistor, both being non-inductive, is connected to a voltage source, the current in the circuit cannot be determined by using Ohm's law. Statement (ID: If the current-voltage characteristic of the non-linear resistor is known, the current-voltage characteristic of the series circuit can be obtained by graphical solution. Statement (I) : Soft magnetic materials, both metallic and ceramic are used for making transformers core, whereas, hard magnetic materials both metallic and ceramic are used for making permanent magnets. Statement (ID: Magnetic materials, both metallic and ceramic are classified as soft or hard according to the magnetic hysteresis loop being narrow or broad. VBY-S-ETE (20 — A)

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